

Alex Junior Gomez Saltachin

PhD Student in Mathematics, PUC-Rio (Brazil)

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Research Interests

Arithmetic Geometry, Homological Mirror Symmetry, Periods of Landau–Ginzburg Models, Topological Quantum Field Theory, Modular Forms, Geometric Representation Theory.

Education

- **PhD in Mathematics**, PUC-Rio, Brazil (2023–present). Advisor: Sergey Galkin.
- **MSc in Mathematics**, Pontificia Universidad Católica del Perú (PUCP), 2019. Thesis: *Modular Forms*.
- **BSc in Mathematics (First Class Honors)**, PUCP, 2015. Thesis: *Elliptic Curves: Mordell–Weil, Roth and Siegel Theorems*.

Selected Schools and Programs

- ICTP School & Workshop on Explicit Arithmetic Geometry (2025)
- ICTP String-Math (2024)
- Arizona Winter School (2022, 2023)
- CIMPA ELGA – Latin American School of Algebraic Geometry (2024)
- IHES Summer School on the Langlands Program (2022)
- ICTP–CIMPA Schools in Arithmetic Geometry (2015, 2018)

Talks (Selected)

- *Periods in Homological Mirror Symmetry and Kloosterman Motives*, ICTP, 2025.
- *Mordell–Weil Theorem*, PUC-Rio Friday Seminar, 2024.
- *Karoubian Completion of Triangulated Categories*, PUC-Rio Friday Seminar, 2024.

Teaching Experience

Graduate Instructor at PUC-Rio and PUCP (2013–2026): Linear Algebra, Algebra, Real Analysis, Calculus, Mathematics for Economists.

Academic Service

- Co-founder and organizer of *38 Movimientos Seminar*.
- Co-organizer of *Meetings on Arithmetic and Algebraic Number Theory Seminar*.

Languages

Spanish (native), English (advanced), Portuguese (advanced), basic Chinese and French.

References

- **Sergey Galkin** (PhD Advisor), PUC-Rio — sergey@puc-rio.br
- **Misha Verbitsky**, IMPA — verbit@impa.br
- **Sergei Burkin**, PUC-Rio — sergei.burkin@puc-rio.br
- **Alfredo Poirier** (MSc Advisor), PUCP — apoirie@pucp.edu.pe
- **Rudy Rosas**, PUCP — rudy.rosas@pucp.pe
- **Richard Gonzales**, PUCP — rgonzalesv@pucp.edu.pe
- **Jaime Cuadros**, PUCP — jcuadros@pucp.edu.pe
- **Percy Fernandez**, PUCP — percy.fernandez@pucp.edu.pe